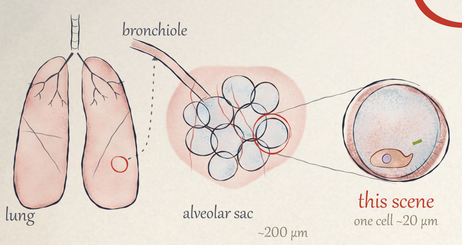
How a macrophage *destroys* what it engulfs

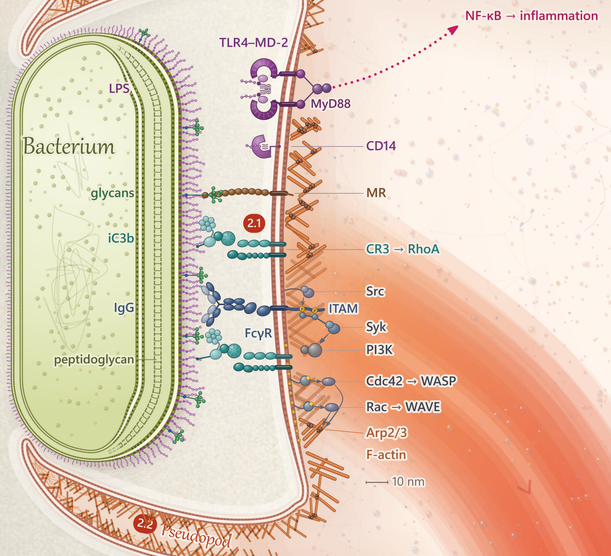
Cell Biology seminar · First-year Medicine
1 Aitor 2 Nerea 2,7 3 Urtzi 4 Eneritz 5 Nora 6 Aitor

Phagocytic route · Enzyme route · Recycling
Membrane fusion · Pathological effect · Inflammatory signal
dashed = possible
Not to scale · molecular detail enlarged
cell ~20 µm · bacterium ~2 µm · receptor ~10 nm

1 Alveolar macrophage

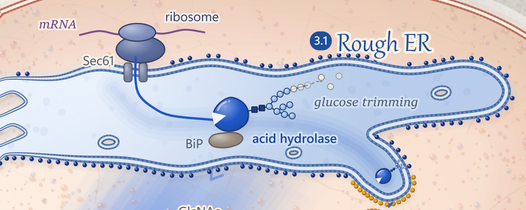


2 Recognition



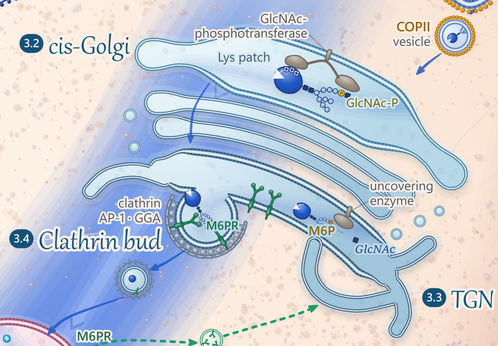
3 Lysosome biogenesis

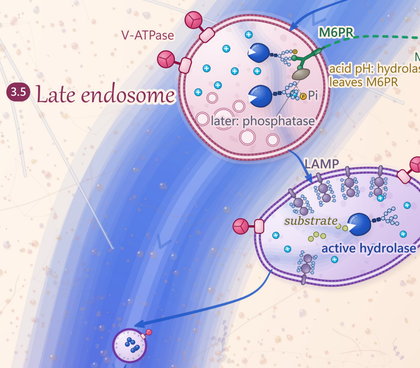
■ GlcNAc ○ mannose ○ glucose ● phosphate



Nucleus

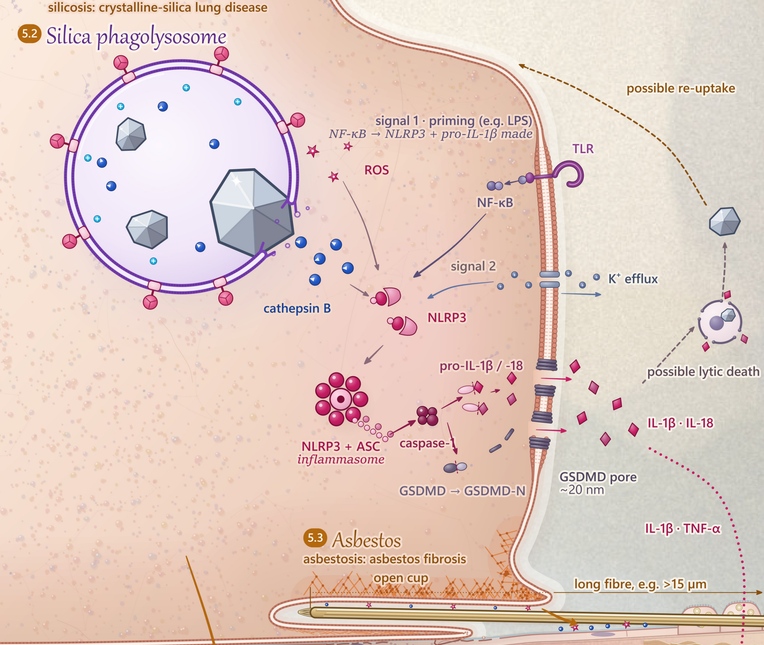
Rough ER

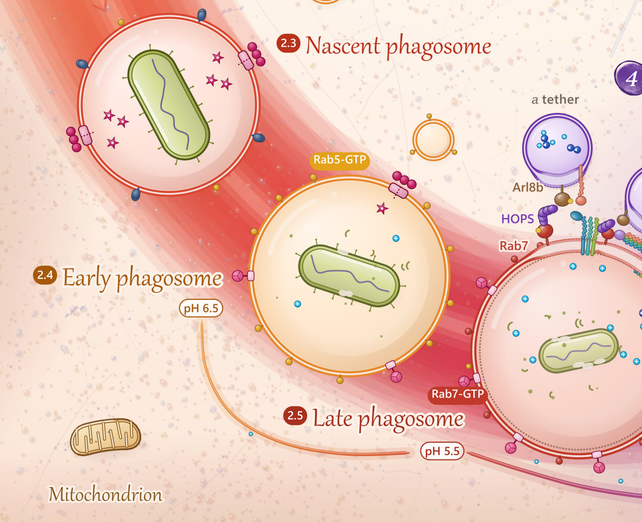


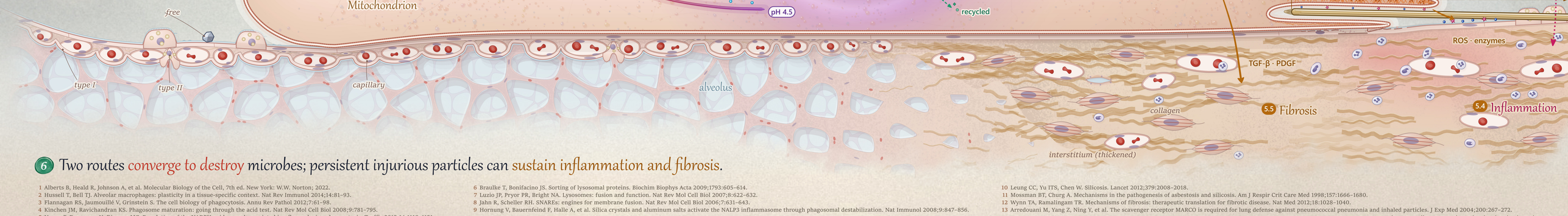


5 Particles









6 Two routes converge to destroy microbes; persistent injurious particles can sustain inflammation and fibrosis.

1 Alberts B, Heald R, Johnson A, et al. Molecular Biology of the Cell, 7th ed. New York: W.W. Norton; 2022.
2 Hussell T, Bell TJ. Alveolar macrophages: plasticity in a tissue-specific context. Nat Rev Immunol 2014;14:81–93.
3 Flannagan RS, Jaumouillé V, Grinstein S. The cell biology of phagocytosis. Annu Rev Pathol 2012;7:61–98.
4 Kinchen JM, Ravichandran KS. Phagosome maturation: going through the acid test. Nat Rev Mol Cell Biol 2008;9:781–795.
5 Nunes P, Demaurex N, Dinauer MC. Regulation of the NADPH oxidase and associated ion fluxes during phagocytosis. Traffic 2013;14:1118–1131.
6 Braulke T, Bonifacino JS. Sorting of lysosomal proteins. Biochim Biophys Acta 2009;1793:605–614.
7 Luzio JP, Pryor PR, Bright NA. Lysosomes: fusion and function. Nat Rev Mol Cell Biol 2007;8:622–632.
8 Jahn R, Scheller RH. SNAREs: engines for membrane fusion. Nat Rev Mol Cell Biol 2006;7:631–643.
9 Hornung V, Bauernfeind F, Halle A, et al. Silica crystals and aluminum salts activate the NALP3 inflammasome through phagosomal destabilization. Nat Immunol 2008;9:847–856.
10 Leung CC, Yu ITS, Chen W. Silicosis. Lancet 2012;379:2008–2018.
11 Mossman BT, Churg A. Mechanisms in the pathogenesis of asbestosis and silicosis. Am J Respir Crit Care Med 1998;157:1666–1680.
12 Wynn TA, Ramalingam TR. Mechanisms of fibrosis: therapeutic translation for fibrotic disease. Nat Med 2012;18:1028–1040.
13 Arredouani M, Yang Z, Ning Y, et al. The scavenger receptor MARCO is required for lung defense against pneumococcal pneumonia and inhaled particles. J Exp Med 2004;200:267–272.

Successive stages in one schematic cell · not to scale; molecular detail enlarged